

Unit 2 Assessment

Knowledge: /14**Application: /20****Thinking: /12****Communication: /16**

1. Fill in the table. **[K2]**

Exponential Form	Logarithmic Form
$2401 = 7^4$	
	$a = \log_b c$

2. Solve for x . Give answer to three decimal places where applicable. **[K2]**

a. $x = \log 37$

b. $x = \log 10\,000$

3. Convert each of the following to exponential form and solve. **[K6]**

a. $n = \log_4 16384$

b. $6 = \log_b 46656$

4. Use the special properties of logarithms to solve the following equations. **[K4]**

a. $M = \log_{11} 11^{21}$

b. $4^{\log_4 57} = x$

c. $x = \log_{23} 78$

5. Evaluate the following using a method of your choice. **[A4]**

a. $\log_9 729 =$

b. $\log_6 7776 - \log_6 36 =$

6. Solve for the unknown variable using a method of your choice. **[A4] [C2]**

a. $5^{11x+23} = 125^{7x}$

b. $2^n = 123$

7. Solve for the unknown variable using a method of your choice. **[T4] [C2]**

$3^{x+5} + 3^x = 177\,876$

8. Describe the transformations on the given logarithmic function. **[C6]**

$$g(x) = -\frac{1}{6}\log_{10}\left[-\frac{1}{6}(x-7)\right] - 85$$

9. Write the equation of a logarithmic function that has been vertically stretched by 5, reflected in the x -axis, horizontally compressed by $1/3$, moved 11 units left and 8 units down. Justify your answer. **[A5] [C2]**

10. The point $(-23, -13)$ is on the transformed function: $y = -2\log_{10}[-5(x + 3)] - 9$. Find the original point on the parent function $f(x) = \log_{10}x$. Justify your answer. **[T4] [C2]**

11. Ethan bought new car worth \$60 000. After 5 years, the car was worth \$35 429.40.
Calculate the depreciation rate of the Ethan's car. **[T4] [C2]**
12. The hydrogen ion concentration of a lemon is about 0.01. What is its pH? **[A3]**
13. How does an earthquake of magnitude 8.4 compare to an earthquake of magnitude 3.7?
Round answer to the nearest whole number. **[A4]**